

PRACTICE PROBLEM SET 2
MTH 210
STATISTICAL COMPUTING

- Suppose U_1, U_2 and U_3 are three independent exponential random variables with means 1, $1/2$ and $1/3$, respectively. If $X = \min\{U_1, U_3\}$ and $Y = \{U_2, U_3\}$, then find
 - (a) $P(X > x, Y > y)$, for $x > 0, y > 0$,
 - (b) if A is a square with the four corner points, $\{(4, 1), (5, 1), (5, 2), (4, 2)\}$, find $P((X, Y) \in A)$
 - How do you approximate using simulation technique?
- Suppose U_1, U_2 and U_3 are three independent identically distributed random variables with the following probability density function:

$$f(x) = \alpha \lambda x^{\alpha-1} e^{-\lambda x^\alpha}; \quad x > 0, \alpha > 0, \lambda > 0.$$

If $X = \min\{U_1, U_3\}$ and $Y = \{U_2, U_3\}$, then find

- (a) $P(X > x, Y > y)$, for $x > 0, y > 0$,
 - (b) if A is a square with the four corner points, $\{(4, 1), (5, 1), (5, 2), (4, 2)\}$, find $P((X, Y) \in A)$
 - How do you approximate $P((X, Y) \in A)$ using simulation technique?
 - Compute $P(X < Y)$.
 - How do you approximate $P(X < Y)$ using simulation technique?
- Let $X_1, X_2, \dots$ be a sequence of independent identically distributed $N(0, 1)$ random variables. Let N be a geometric random variable with the following probability mass function

$$P(N = k) = \frac{1}{2^k}; \quad k = 1, 2, \dots$$

Suppose X_i 's are N are independently distributed. If $Y = \sum_{i=1}^N X_i$, then

- Find the probability density function of Y .
 - Find $E(Y)$.
 - How do you approximate $E(Y)$ using simulation technique?
- Suppose X is a random variable with the following density function

$$f(x) = \frac{1}{2}(1+x)e^{-x}; \quad x > 0.$$

- How do you generate sample from the above density function using acceptance-rejection method?
- How do you generate sample from the above density function using ratio of uniform method?
- How do you compute $E(\ln X)$?

5. Suppose (X, Y) is a random vector with the following joint probability density function

$$f(x, y) = ce^{-x-y-xy}; \quad x > 0, y > 0$$

- How will you compute c based on simulation technique?
- How will you compute $E(XY)$ based on simulation technique?

6. Estimate $\int_0^1 e^x dx$ using importance sampling technique.

7. Estimate $\int_0^\infty e^{-x^2} dx$ using importance sampling technique.

8. Estimate different moments of a gamma random variable using importance sampling technique.